Introduction

The laboratory work is intended to give students the opportunity to put the material in this module into practice and to gain a critical understanding of the models presented in the lectures. The coursework component of the module consists of five work sheets, to be done in your own time as there are no scheduled laboratory classes. By necessity, these models are somewhat simplistic, as the module covers a broad range of subjects. A critical understanding of the models means not only knowing what inputs and outputs are relevant to each model, but also being able to comment on the relevance and accuracy of the output. Marks will be awarded for all these aspects.

The results, discussion and conclusions arising from this and subsequent exercises should be written in an electronic lab-book (MS Word or similar), along with any graphical output from the programmes. This is not a formal report, but it should still be clearly laid out and all variables should be defined. The results should be in sentence form, rather than just cut and pasted from the Matlab window. For example, the following is inadequate:

*'>> d=sqrt(2*a*ht)*
d =

99.5359'

and should be replaced by:

*'The horizon distance was calculated from the formula d=sqrt(2*a*ht), where a is the earth radius in km and ht is the antenna height in km. It was found to be 99.5 km.'*

Matlab code should not be put in the lab-book. If possible, please convert the finished lab-book to pdf format before uploading it to the EG4217/EG7217 Blackboard site. The mark for the lab-book contributes 30% of the total marks available for the module.

You must upload the document by **12.00 noon on Friday 27th November 2022**. Late submissions will lose 10% of the total possible mark for the first 24 hours or part thereof, and an additional 5% for each subsequent 24 hour period or part thereof.

You are reminded that while the outputs of the software will be the same for many students, the discussions and conclusions must be **your own work**. All work that is not your own MUST be referenced in your logbook. When you submit your work you **must** include the following paragraph at the beginning of your lab-book:

I declare that this assignment is my own work, that sources of reference are acknowledged and that it has not been submitted for any other course. I understand that plagiarism is a serious offence under the University's regulations and that appropriate penalties will be applied if I am found to have submitted plagiarised work.

The software

The Matlab programs are available from the module Blackboard site. They do not need to be altered or edited. To run a Matlab model, download it from the Blackboard site and place it in a newly made directory called, for example, "AdvComCoursework", which should be on your Z-drive (your personal space on the UoL computer system that can be accessed from any computer on campus). Then open Matlab and navigate to the AdvComCoursework directory. You should see the programme in the part of the Matlab window called "Current Folder".

Now move your cursor to the part of the window marked "Command Window" You should see the prompt ">>". Type the name of the program, followed by round brackets and the value of any input parameters (as numbers), separated by commas. If you're unsure what the input parameters are, then double-click on the programme in "Current Folder" and the programme's

code will appear in the part of the window marked "Editor". The first line of code will tell you the required parameters and their correct order.

For example, to run freespace.m type `freespace(1000,50)`

This will return the calculated value of free-space loss for a 1000 MHz radio wave that travels 50km.

Some additional Matlab code may need to be written to present the results in the most appropriate format (see below). Many of the Matlab functions accept one of the input parameters in vector form, allowing the output parameter to be plotted in one line, e.g.

```
plot([1:20],freespace(1000,[1:20]))
```

will plot out the freespace loss from 1 to 20km in 1km steps for a 1GHz signal. A slower, but a more C- or Fortran-like, alternative is

```
for range=1:20
loss(range)=freespace(1000,range);
end
plot([1:20],loss,'k') %'k' means draw line in black
```

All of the functions are sufficiently well documented by comments to explain what they do and what inputs are required. You should have a reasonable grasp what each model does before running the code. Please be careful to look at the units of each input variable and then use appropriate values. A few extra Matlab commands may be useful (look up the details using the online help, e.g. at the command line type help subplot):

- hold on : enables multiple lines to be drawn on the same axes
- subplot : can be used to generate multiple axes on the same piece of paper
- orient landscape (or tall) : sets the paper in the given orientation
- title, xlabel and ylabel : add the appropriate axis labels and title to your graph
- print –dtiff <filename.tif> : saves the current figure in tiff format

(which can be imported into word).

- plot : see the help for details of how to add plot symbols, introduce dashed lines, etc.

Coursework 1

Task 1

freespace.m

This programme calculates free space loss assuming isotropic antennas. We shall investigate the effect of frequency and distance on the loss.

Plot the free-space loss for ranges between 1 km and 60 km for radio frequencies 100 MHz and 1 GHz.

To make this curve easy for the reader to view and understand:
Plot these curves in different colours on the same graph using 'hold on'.
Add a graticule using the command 'grid on'
Add a legend using the command 'legend'.
Add axis labels using 'xlabel' and 'ylabel', not forgetting units.
Add a figure number and caption to the plot
Add some text, referring to Figure 1 and describing the trend.
Explain why loss increases with distance.
(Subsequent outputs should also all have these embellishments to make them clear.)

Task 2

planereflec2.m

This function calculates the loss associated with propagation over a plane earth using the two-ray ground model and assuming isotropic antennas. Please Note: This programme requires all distances in km, as can be seen by double-clicking on the programme in the Matlab 'Current Folder' pane and examining the code in the Editor pane.

a) Plot the loss as a function of Tx-Rx distance for distances between 10km and 100 km, transmit and receive antennas of height 80 m and frequencies 500 MHz and 1 GHz.

b) What physical factors limit the values of range for which the model can be used?
c) Explain why the loss values form peaks at certain distances

Task 3

No Matlab programme needed

Plot a graph of the maximum distance for which two antennas of equal height can maintain a line-of-sight link, assuming a smooth spherical earth. The graph should show maximum distance in km as a function of antenna height in m for antennas of height 5 to 100m. Use the formula d = sqrt(2* R* ht) + sqrt(2* R* hr), where ht and hr are the heights of the transmit and receive antennas respectively and R is the effective earth radius, as shown in the lecture.

Task 4

p526.m

This programme implements the model found in ITU Recommendation P526 (on the Bb site) for propagation loss beyond the radio horizon. It outputs the extra loss in dB due to diffraction of the radio signal over the curvature of the earth for over-the-horizon radio links.

Plot the diffraction loss predicted by this model as a function of distance for a vertically polarised radio wave of frequency 6 GHz travelling over the sea and antenna heights of 100m. How much diffraction loss is caused at the maximum LOS distance ?

Task 5

Using all the programmes mentioned above

Figure 1 shows the Channel Islands, which are situated in the English Channel between the UK and France. An omnidirectional antenna with 2dB of gain transmits vertically polarised radio waves with a power of 30 watts and a frequency of 500 MHz from the highest point on the island of Jersey (150 m above the mean sea level). Jersey is marked as the dot next to the letter J in Figure 1. The transmitter broadcasts a radio service to Guernsey (marked G in Figure 1), Alderney (marked A) , Portland (P) and Leicester (L). One more receiver (not shown on the map) is placed at location J2, in sight of the transmitter antenna at 300 m distance for diagnostic purposes. Between it and the transmitter antenna the ground is hilly and tree covered.
All receive antennas have 2 dB gain. Gains are not included in the loss values given in the table.

Complete the following table using the three programmes already introduced:
L1 = loss calculated using freespace.m,
L2 = loss calculated using planereflec2.m,
L3 = loss calculated using p526.m,

In the last column, consider the three different loss estimates given by the models, and choose which one you think applies best to the path in question (L1, L2 or L3). Then do a power budget calculation using this value to get the value of Received power (Best Estimate) in dBW.
State reasoning for each case. High precision is not expected in this exercise.
Please remember that the output of planereflec.m includes free-space loss, whereas p526.m does not.

Receiver Location	Distance from Jersey(km)	Receive Antenna height (m)	L1	L2	L3	Received power dBW(Best Estimate)
Guernsey	33	10				
Alderney	50			125		
Location J2	0.3	1				
Portland		10	130			
Leicester		30				

In this task, you should assume
1 degree of latitude = 111km
1 degree of longitude = 72 km at this latitude

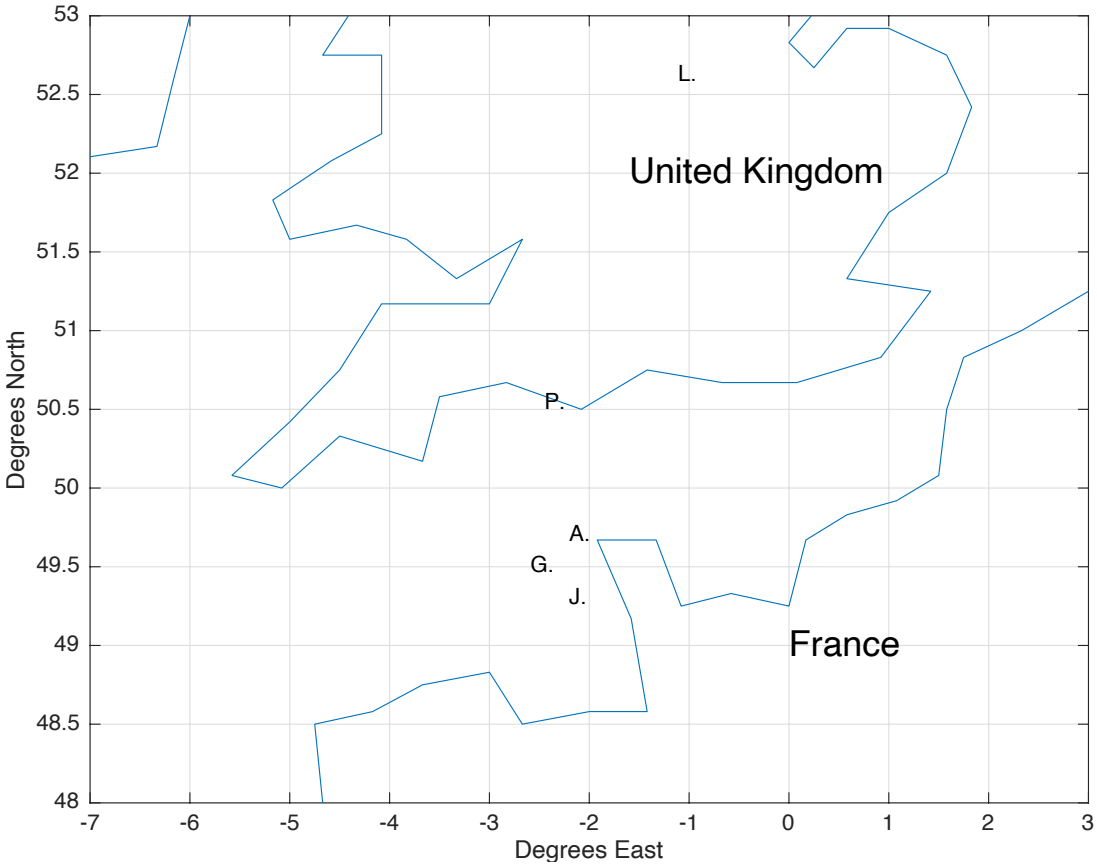


Figure 1 The Channel Islands

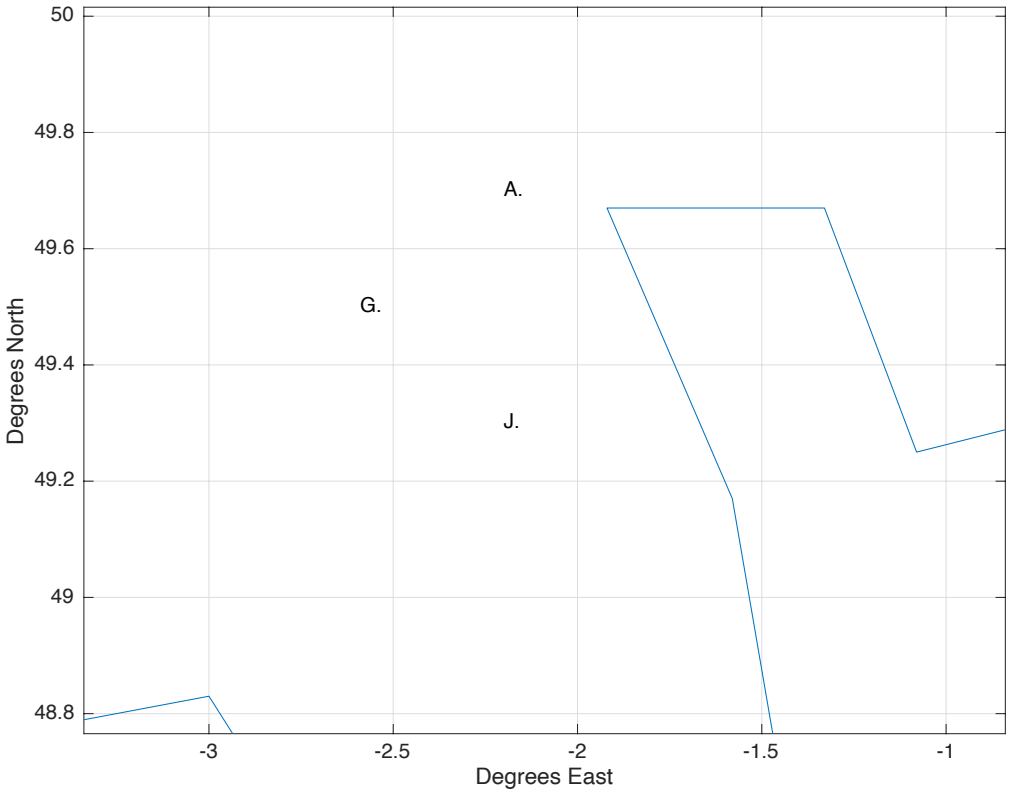


Fig 2 Inset of Figure 1

EG4217/EG7217 Advanced Communications

Coursework 4

Task 1

ITUdryair.m, sat_dist.m and **freespace.m**

The programme ITUdryair.m calculates the loss due to absorption by oxygen.
The programme sat_dist calculates the distance from a satellite to the ground-station.

A communications satellite orbits at a height of 500 km above the earth. It transmits a signal with an EIRP of 100W at a carrier frequency of 52 GHz to a ground-station at sea-level with a gain of 30 dBi. The threshold value for the received signal is –130 dBW.
The air pressure measured at the ground-station is 1013 hPa and at 12 km height, it is measured as 150 hPa. The temperature at the ground-station is 17°C.

a. Calculate the scale height for oxygen pressure.
b. Use ITUdryair.m and this scale height to calculate the oxygen loss when the satellite is overhead.
c. Divide by the loss in part (b) by a vector containing the sines of elevation angles [5,10,15,…,90] degrees to get the loss for these elevations and plot the loss as a function the elevation angle.
d. Calculate the distance of the satellite from the ground station for this same range of elevation angles
e. Add free-space loss to the oxygen loss to get the total loss as a function the elevation angle
f. Use the link-budget to calculate the minimum elevation angle at which the signal from the satellite can be received.

Task 2

SAMrain.m

This function calculates the loss associated with rain; the former for 0.01% of the time

A geostationary satellite has an elevation of 20° from Leicester, which has an altitude of 62 m above mean sea-level and it has approximately the same elevation as seen from a few miles away in Buxton, Derbyshire, which has an altitude of 330 m. Use the approximate latitude of 53°N for both stations and calculate how much better or worse the signal would be in Buxton, compared to Leicester, for the worst 52 minutes of a typical year. The satellite's carrier signal has a frequency of 35 GHz.

Task 3

ITUrainattenuation.m

From document ITU-R P837, find the rain rate in mm/hour for the rainiest 0.01% of the time in Kolkata (Calcutta), India.
A geostationary TV satellite has an elevation of 64°, as seen from Kolkata, which is at sea-level. A 15 GHz TV signal is received by a large antenna and distributed to subscribers. In dry weather, the signal-to-noise ratio at the subscribers' homes is 10 dB, which is sufficient to ensure a good TV signal. However, the operators are worried that if it goes below 5dB, viewers will complain. How many hours per year would you expect to see this level of degradation? *(To help you answer this question, use the inbuilt Matlab command p=logspace(-1,-3,10) to create a range of percentages. These percentages may be input to ITUrainattenuation.m as the last input parameter. Then use the loglog command to plot rain attenuation against p on a log-log graph.)*

What other effect (that is not taken into consideration in the Matlab programme) might make this value inaccurate?

Task 4

ITUxpd.m

This programme implements the model found in ITU Recommendation P618 to determine the cross-polar discrimination seen when signals are sent simultaneously on different polarisations of the same wave.

A 15 GHz radio wave with left-handed circular polarisation is beamed from ground-station to a satellite at 20^0 elevation. When the weather at the ground station is fine, the satellite receives the signal with an signal-to-noise ratio of 25 dB, but in the worst 0.01% of times, the SNR reduces to 20 dB. To increase throughput, the satellite operator decides to duplex two signals together; one with left-handed circular polarisation and one with right-handed circular polarisation.

a. What will the value of the cross-polar discrimination for these signals?
b. What will be the SINR (the signal-to-(interferer plus noise)-ratio) for this signal for 0.01% of the time?